Theoretical Physics 2 - Blatt 03

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Section 1: Moments of statistical distributions Compute the expectation value $\mu = \langle X \rangle$ and the variance

$$\sigma^2 = \langle (X-\mu)^2 \rangle$$

for the given distributions. In part (c), also compute the third central moment

$$\mu_3 = \langle (X-\mu)^3 \rangle.$$

(a) Bernoulli distribution with

$$P(X) = p^X (1-p)^{1-X}, \qquad X \in \{0,1\}.$$

Calculation

Since X only takes the values 0 and 1, we have

$$P(X=0) = 1-p, \qquad P(X=1) = p.$$

Therefore,

$$\begin{aligned} \mu = \langle X \rangle &= \sum_{X \in \{0,1\}} X\, P(X) \\ &= 0 \cdot (1-p) + 1 \cdot p \\ &= p. \end{aligned}$$

For the variance, we may either use $\sigma^2 = \langle X^2 \rangle - \langle X \rangle^2$ or directly the definition. Since $X^2 = X$ for $X \in \{0,1\}$, we get

$$\langle X^2 \rangle = \langle X \rangle = p.$$

Hence

$$\begin{aligned} \sigma^2 &= \langle X^2 \rangle - \mu^2 \\ &= p - p^2 \\ &= p(1-p). \end{aligned}$$

Answer

For the Bernoulli distribution,

$$\boxed{\mu = p, \qquad \sigma^2 = p(1-p).}$$

(b) Poisson distribution with

$$P_\lambda(X) = \frac{\lambda^X e^{-\lambda}}{X!}, \qquad X \in \mathbb{N}.$$

Calculation

We compute the expectation value:

$$\mu = \langle X \rangle = \sum_{X=0}^{\infty} X \frac{\lambda^X e^{-\lambda}}{X!}.$$

The term $X = 0$ vanishes, so

$$\begin{aligned}
\mu &= e^{-\lambda} \sum_{X=1}^{\infty} X \frac{\lambda^X}{X!} \\
&= e^{-\lambda} \sum_{X=1}^{\infty} \frac{\lambda^X}{(X-1)!} \\
&= \lambda e^{-\lambda} \sum_{X=1}^{\infty} \frac{\lambda^{X-1}}{(X-1)!}.
\end{aligned}$$

Set $n = X - 1$. Then

$$\begin{aligned}
\mu &= \lambda e^{-\lambda} \sum_{n=0}^{\infty} \frac{\lambda^n}{n!} \\
&= \lambda e^{-\lambda} e^{\lambda} \\
&= \lambda.
\end{aligned}$$

Next, compute $\langle X(X-1) \rangle$:

$$\langle X(X-1) \rangle = \sum_{X=0}^{\infty} X(X-1) \frac{\lambda^X e^{-\lambda}}{X!}.$$

Again only $X \geq 2$ contributes:

$$\begin{aligned}
\langle X(X-1) \rangle &= e^{-\lambda} \sum_{X=2}^{\infty} \frac{\lambda^X}{(X-2)!} \\
&= \lambda^2 e^{-\lambda} \sum_{X=2}^{\infty} \frac{\lambda^{X-2}}{(X-2)!}.
\end{aligned}$$

With $m = X - 2$:

$$\begin{aligned}
\langle X(X-1) \rangle &= \lambda^2 e^{-\lambda} \sum_{m=0}^{\infty} \frac{\lambda^m}{m!} \\
&= \lambda^2.
\end{aligned}$$

Since

$$X^2 = X(X-1) + X,$$

we obtain

$$\begin{aligned}\langle X^2\rangle &= \langle X(X-1)\rangle + \langle X\rangle \\ &= \lambda^2 + \lambda.\end{aligned}$$

Therefore,

$$\begin{aligned}\sigma^2 &= \langle X^2\rangle - \mu^2 \\ &= (\lambda^2+\lambda) - \lambda^2 \\ &= \lambda.\end{aligned}$$

Answer

For the Poisson distribution,

$$\boxed{\mu = \lambda, \qquad \sigma^2 = \lambda.}$$

Thus for a Poisson distribution the mean and variance are equal.

(c) Gaussian distribution with density

$$P(X) = \frac{1}{\sqrt{2\pi b}} \exp\left[-\frac{(X-a)^2}{2b}\right], \qquad X,a,b \in \mathbb{R},$$

with $b > 0$.

Calculation

This is the normal distribution with mean parameter a and variance parameter b. To see this explicitly, introduce

$$y = X - a.$$

Then

$$P(X)\,dX = \frac{1}{\sqrt{2\pi b}} e^{-y^2/(2b)}\,dy,$$

which is an even function of y.
First moment:

$$\begin{aligned}\mu = \langle X\rangle &= \int_{-\infty}^{\infty} X\,P(X)\,dX \\ &= \int_{-\infty}^{\infty} (a+y)\frac{1}{\sqrt{2\pi b}} e^{-y^2/(2b)}\,dy \\ &= a\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi b}} e^{-y^2/(2b)}\,dy + \int_{-\infty}^{\infty} y\frac{1}{\sqrt{2\pi b}} e^{-y^2/(2b)}\,dy.\end{aligned}$$

The first integral is 1 and the second vanishes because the integrand is odd. Hence

$$\mu = a.$$

The variance is

$$\begin{aligned}\sigma^2 &= \langle (X-\mu)^2 \rangle \\ &= \langle (X-a)^2 \rangle \\ &= \int_{-\infty}^{\infty} y^2 \frac{1}{\sqrt{2\pi b}} e^{-y^2/(2b)}\, dy.\end{aligned}$$

This is the standard second central moment of a Gaussian and equals b. Hence

$$\sigma^2 = b.$$

The third central moment is

$$\begin{aligned}\mu_3 &= \langle (X-\mu)^3 \rangle \\ &= \langle (X-a)^3 \rangle \\ &= \int_{-\infty}^{\infty} y^3 \frac{1}{\sqrt{2\pi b}} e^{-y^2/(2b)}\, dy.\end{aligned}$$

Since the integrand is odd and the integration interval is symmetric,

$$\mu_3 = 0.$$

This expresses the symmetry of the Gaussian distribution about its mean.

Answer

For the Gaussian distribution,

$$\boxed{\mu = a, \qquad \sigma^2 = b, \qquad \mu_3 = 0.}$$

Section 2: Charged 1D harmonic oscillator in a uniform electric field We consider the Hamiltonian

$$H(E) = \frac{P^2}{2m} + \frac{1}{2} m\omega^2 X^2 - eEX.$$

Determine the energy eigenvalues $E_n(E)$ and eigenfunctions $\phi_n(x)$, discuss the energy shift and wavefunctions, and compute the expectation value of the electric dipole moment

$$\langle D \rangle = e(\phi_n, X\phi_n).$$

(a) Energy spectrum, shifted coordinate, eigenfunctions, and dipole moment.

Calculation

The key step is to complete the square in the potential:

$$\begin{aligned}\frac{1}{2}m\omega^2x^2 - eEx &= \frac{1}{2}m\omega^2\left(x^2 - 2\frac{eE}{m\omega^2}x\right)\\ &= \frac{1}{2}m\omega^2\left(x - \frac{eE}{m\omega^2}\right)^2 - \frac{1}{2}m\omega^2\left(\frac{eE}{m\omega^2}\right)^2\\ &= \frac{1}{2}m\omega^2\left(x - \frac{eE}{m\omega^2}\right)^2 - \frac{e^2E^2}{2m\omega^2}.\end{aligned}$$

Define the shifted coordinate

$$u = x - \frac{eE}{m\omega^2}.$$

Then the Hamiltonian becomes

$$H(E) = \frac{P^2}{2m} + \frac{1}{2}m\omega^2u^2 - \frac{e^2E^2}{2m\omega^2}.$$

Since $u = X - \frac{eE}{m\omega^2}$ differs from X only by a constant,

$$[u,P] = [X,P] = i\hbar.$$

Thus u and P are again canonically conjugate variables.
Therefore the problem is exactly the usual harmonic oscillator in the variable u, plus a constant energy shift. For the ordinary oscillator, the eigenvalues are

$$E_n^{(0)} = \hbar\omega\left(n + \frac{1}{2}\right), \qquad n = 0,1,2,\ldots$$

Hence here

$$\boxed{E_n(E) = \hbar\omega\left(n + \frac{1}{2}\right) - \frac{e^2E^2}{2m\omega^2}}.$$

The eigenfunctions are the usual oscillator eigenfunctions, but centered at

$$x_0 = \frac{eE}{m\omega^2}.$$

Thus

$$\phi_n(x;E) = \varphi_n\left(x - \frac{eE}{m\omega^2}\right),$$

where φ_n is the n-th eigenfunction of the standard harmonic oscillator:

$$\varphi_n(u) = \frac{1}{\sqrt{2^n n!}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4} H_n\left(\sqrt{\frac{m\omega}{\hbar}}\,u\right)\exp\left(-\frac{m\omega}{2\hbar}u^2\right).$$

Therefore

$$\boxed{\phi_n(x;E) = \frac{1}{\sqrt{2^n n!}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4} H_n\left(\sqrt{\frac{m\omega}{\hbar}}\left(x - \frac{eE}{m\omega^2}\right)\right)\exp\left[-\frac{m\omega}{2\hbar}\left(x - \frac{eE}{m\omega^2}\right)^2\right]}.$$

The energy shift is the same for all levels:

$$\Delta E = -\frac{e^2E^2}{2m\omega^2}.$$

Hence the level spacing remains unchanged:

$$E_{n+1}(E) - E_n(E) = \hbar\omega.$$

Now compute the dipole moment:

$$\langle D\rangle = e\langle X\rangle.$$

Since the eigenfunctions are centered at $x_0 = \frac{eE}{m\omega^2}$ and the standard oscillator eigenstates satisfy $\langle u\rangle_n = 0$, we have

$$\langle X\rangle_n = \left\langle u + \frac{eE}{m\omega^2}\right\rangle_n = \langle u\rangle_n + \frac{eE}{m\omega^2} = \frac{eE}{m\omega^2}.$$

Thus

$$\boxed{\langle D\rangle_n = e\langle X\rangle_n = \frac{e^2E}{m\omega^2}}.$$

This expectation value is independent of n.

Answer

The electric field shifts the harmonic oscillator potential rigidly by

$$x_0 = \frac{eE}{m\omega^2},$$

without changing its curvature. Therefore all energy levels are shifted by the same constant:

$$\boxed{E_n(E) = \hbar\omega\left(n + \frac{1}{2}\right) - \frac{e^2E^2}{2m\omega^2}.}$$

The eigenfunctions are the usual harmonic oscillator wavefunctions translated by x_0, and the dipole moment expectation value is

$$\boxed{\langle D\rangle_n = \frac{e^2E}{m\omega^2}.}$$

Section 3: Coherent states of the 1D harmonic oscillator — properties Let

$$\phi_\alpha := e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \varphi_n, \qquad \alpha \in \mathbb{C},$$

where φ_n are the harmonic oscillator energy eigenstates,

$$\hat{H}\varphi_n = \hbar\omega \left(n + \frac{1}{2}\right) \varphi_n, \qquad \hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right).$$

(a) Show that coherent states are eigenstates of the annihilation operator:

$$\hat{a}\phi_\alpha = \alpha\phi_\alpha.$$

Calculation

We use

$$\hat{a}\varphi_n = \sqrt{n}\,\varphi_{n-1}, \qquad \hat{a}\varphi_0 = 0.$$

Then

$$\begin{aligned}
\hat{a}\phi_\alpha &= e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \hat{a}\varphi_n \\
&= e^{-|\alpha|^2/2} \sum_{n=1}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \sqrt{n}\,\varphi_{n-1} \\
&= e^{-|\alpha|^2/2} \sum_{n=1}^{\infty} \frac{\alpha^n}{\sqrt{(n-1)!}} \varphi_{n-1}.
\end{aligned}$$

Set $m = n - 1$:

$$\begin{aligned}
\hat{a}\phi_\alpha &= e^{-|\alpha|^2/2} \sum_{m=0}^{\infty} \frac{\alpha^{m+1}}{\sqrt{m!}} \varphi_m \\
&= \alpha e^{-|\alpha|^2/2} \sum_{m=0}^{\infty} \frac{\alpha^m}{\sqrt{m!}} \varphi_m \\
&= \alpha\phi_\alpha.
\end{aligned}$$

Answer

$$\boxed{\hat{a}\phi_\alpha = \alpha\phi_\alpha.}$$

Thus ϕ_α is indeed an eigenstate of the annihilation operator with eigenvalue α.

(b) Show that coherent states are normalized: $\|\phi_\alpha\| = 1$.

Solution

Using orthonormality $(\varphi_n,\varphi_m) = \delta_{nm}$,

$$\begin{aligned}
\|\phi_\alpha\|^2 &= (\phi_\alpha,\phi_\alpha) \\
&= e^{-|\alpha|^2} \sum_{n,m=0}^{\infty} \frac{(\alpha^*)^m \alpha^n}{\sqrt{m!n!}} (\varphi_m,\varphi_n) \\
&= e^{-|\alpha|^2} \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} \\
&= e^{-|\alpha|^2} e^{|\alpha|^2} \\
&= 1.
\end{aligned}$$

Hence

$$\boxed{\|\phi_\alpha\| = 1.}$$

(c) Show completeness:

$$\frac{1}{\pi} \int_{\mathbb{C}} d^2\alpha\, \phi_\alpha (\phi_\alpha,\psi) = \psi \qquad \forall \psi \in L^2(\mathbb{R}).$$

Calculation

It is sufficient to show that the operator

$$\hat{I}_c := \frac{1}{\pi} \int_{\mathbb{C}} d^2\alpha\, |\phi_\alpha\rangle\langle\phi_\alpha|$$

acts as the identity on the number basis $\{\varphi_n\}$.
Insert the coherent-state expansion:

$$|\phi_\alpha\rangle\langle\phi_\alpha| = e^{-|\alpha|^2} \sum_{n,m=0}^{\infty} \frac{\alpha^n (\alpha^*)^m}{\sqrt{n!m!}} |\varphi_n\rangle\langle\varphi_m|.$$

Therefore

$$\hat{I}_c = \frac{1}{\pi} \sum_{n,m=0}^{\infty} \frac{|\varphi_n\rangle\langle\varphi_m|}{\sqrt{n!m!}} \int_{\mathbb{C}} d^2\alpha\, e^{-|\alpha|^2} \alpha^n (\alpha^*)^m.$$

Using polar coordinates $\alpha = re^{i\theta}$, one gets the standard integral

$$\frac{1}{\pi} \int_{\mathbb{C}} d^2\alpha\, e^{-|\alpha|^2} \alpha^n (\alpha^*)^m = n!\, \delta_{nm}.$$

Hence

$$\begin{aligned}\hat{I}_c &= \sum_{n,m=0}^{\infty} \frac{n!\,\delta_{nm}}{\sqrt{n!m!}} |\varphi_n\rangle\langle\varphi_m| \\ &= \sum_{n=0}^{\infty} |\varphi_n\rangle\langle\varphi_n| \\ &= \mathbb{1}.\end{aligned}$$

Therefore,

$$\frac{1}{\pi}\int_{\mathbb{C}} d^2\alpha\, \phi_\alpha(\phi_\alpha,\psi) = \psi.$$

Answer

The coherent states are overcomplete and satisfy the resolution of the identity

$$\boxed{\frac{1}{\pi}\int_{\mathbb{C}} d^2\alpha\, |\phi_\alpha\rangle\langle\phi_\alpha| = \mathbb{1}.}$$

(d) Show that the probability to measure the energy

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right)$$

in the state ϕ_α is Poisson distributed.

Solution

The amplitude for the level n is

$$\langle\varphi_n|\phi_\alpha\rangle = e^{-|\alpha|^2/2}\frac{\alpha^n}{\sqrt{n!}}.$$

Therefore the probability is

$$\begin{aligned}P_n &= |\langle\varphi_n|\phi_\alpha\rangle|^2 \\ &= e^{-|\alpha|^2}\frac{|\alpha|^{2n}}{n!}.\end{aligned}$$

This is precisely a Poisson distribution with parameter

$$\lambda = |\alpha|^2.$$

Answer

$$\boxed{P_n = e^{-|\alpha|^2}\frac{|\alpha|^{2n}}{n!}.}$$

Thus the energy measurement statistics in a coherent state are Poissonian.

(e) Show that the mean energy and energy uncertainty are

$$\langle H\rangle_\alpha = \hbar\omega\left(|\alpha|^2 + \frac{1}{2}\right), \qquad \Delta H_\alpha = \hbar\omega|\alpha|.$$

Calculation

Since

$$\hat{H} = \hbar\omega\left(\hat{N} + \frac{1}{2}\right), \qquad \hat{N} = \hat{a}^\dagger\hat{a},$$

it is enough to compute the moments of $\hat{N}$.
Because $\hat{a}\phi_\alpha = \alpha\phi_\alpha$, we get

$$\begin{aligned}\langle\hat{N}\rangle_\alpha &= \langle\phi_\alpha|\hat{a}^\dagger\hat{a}|\phi_\alpha\rangle \\ &= \langle\hat{a}\phi_\alpha|\hat{a}\phi_\alpha\rangle \\ &= |\alpha|^2.\end{aligned}$$

Hence

$$\langle H\rangle_\alpha = \hbar\omega\left(\langle\hat{N}\rangle_\alpha + \frac{1}{2}\right) = \hbar\omega\left(|\alpha|^2 + \frac{1}{2}\right).$$

Next,

$$\Delta H_\alpha = \hbar\omega\,\Delta N_\alpha.$$

So we compute

$$\Delta N_\alpha^2 = \langle\hat{N}^2\rangle_\alpha - \langle\hat{N}\rangle_\alpha^2.$$

Since the number distribution is Poisson with parameter $|\alpha|^2$, its variance is also $|\alpha|^2$. Therefore

$$\Delta N_\alpha^2 = |\alpha|^2 \qquad \Rightarrow \qquad \Delta N_\alpha = |\alpha|.$$

Hence

$$\boxed{\Delta H_\alpha = \hbar\omega|\alpha|.}$$

The relative energy fluctuation behaves as

$$\frac{\Delta H_\alpha}{\langle H\rangle_\alpha} = \frac{|\alpha|}{|\alpha|^2 + \frac{1}{2}} \sim \frac{1}{|\alpha|} \qquad (|\alpha| \gg 1),$$

so it vanishes in the quasi-classical limit.

Answer

$$\boxed{\langle H\rangle_\alpha = \hbar\omega\left(|\alpha|^2 + \frac{1}{2}\right), \qquad \Delta H_\alpha = \hbar\omega|\alpha|.}$$

(f) Show that the position and momentum expectation values and uncertainties are

$$\langle X\rangle_\alpha = \sqrt{\frac{2\hbar}{m\omega}}\Re(\alpha), \qquad \langle P\rangle_\alpha = \sqrt{2m\hbar\omega}\Im(\alpha),$$

and

$$\Delta X_\alpha = \sqrt{\frac{\hbar}{2m\omega}}, \qquad \Delta P_\alpha = \sqrt{\frac{m\hbar\omega}{2}}.$$

Calculation

We use the operator identities

$$X = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger), \qquad P = -i\sqrt{\frac{m\hbar\omega}{2}}(\hat{a} - \hat{a}^\dagger).$$

Therefore

$$\begin{aligned}
\langle X\rangle_\alpha &= \sqrt{\frac{\hbar}{2m\omega}}\left(\langle\hat{a}\rangle_\alpha + \langle\hat{a}^\dagger\rangle_\alpha\right) \\
&= \sqrt{\frac{\hbar}{2m\omega}}(\alpha + \alpha^*) \\
&= \sqrt{\frac{2\hbar}{m\omega}}\Re(\alpha).
\end{aligned}$$

Similarly,

$$\begin{aligned}
\langle P\rangle_\alpha &= -i\sqrt{\frac{m\hbar\omega}{2}}\left(\langle\hat{a}\rangle_\alpha - \langle\hat{a}^\dagger\rangle_\alpha\right) \\
&= -i\sqrt{\frac{m\hbar\omega}{2}}(\alpha - \alpha^*) \\
&= \sqrt{2m\hbar\omega}\Im(\alpha).
\end{aligned}$$

Now for the variances. First,

$$X^2 = \frac{\hbar}{2m\omega}\left(\hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} + (\hat{a}^\dagger)^2\right).$$

Using $\hat{a}\hat{a}^\dagger = \hat{a}^\dagger\hat{a} + 1$ and the coherent-state expectation values

$$\langle\hat{a}\rangle = \alpha, \quad \langle\hat{a}^\dagger\rangle = \alpha^*, \quad \langle\hat{a}^2\rangle = \alpha^2, \quad \langle(\hat{a}^\dagger)^2\rangle = (\alpha^*)^2, \quad \langle\hat{a}^\dagger\hat{a}\rangle = |\alpha|^2,$$

we find

$$\begin{aligned}
\langle X^2\rangle_\alpha &= \frac{\hbar}{2m\omega}\left(\alpha^2 + (\alpha^*)^2 + |\alpha|^2 + (|\alpha|^2 + 1)\right) \\
&= \frac{\hbar}{2m\omega}\left(\alpha^2 + (\alpha^*)^2 + 2|\alpha|^2 + 1\right).
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\langle X\rangle_\alpha^2 &= \frac{\hbar}{2m\omega}(\alpha + \alpha^*)^2 \\
&= \frac{\hbar}{2m\omega}\left(\alpha^2 + (\alpha^*)^2 + 2|\alpha|^2\right).
\end{aligned}$$

Therefore

$$\Delta X_\alpha^2 = \langle X^2\rangle_\alpha - \langle X\rangle_\alpha^2 = \frac{\hbar}{2m\omega},$$

so

$$\boxed{\Delta X_\alpha = \sqrt{\frac{\hbar}{2m\omega}}.}$$

Likewise one finds

$$\boxed{\Delta P_\alpha = \sqrt{\frac{m\hbar\omega}{2}}.}$$

Hence

$$\Delta X_\alpha \Delta P_\alpha = \sqrt{\frac{\hbar}{2m\omega}}\sqrt{\frac{m\hbar\omega}{2}} = \frac{\hbar}{2}.$$

Thus coherent states are minimum-uncertainty states.

Answer

$$\boxed{\langle X\rangle_\alpha = \sqrt{\frac{2\hbar}{m\omega}}\Re(\alpha), \qquad \langle P\rangle_\alpha = \sqrt{2m\hbar\omega}\Im(\alpha)}$$

and

$$\boxed{\Delta X_\alpha = \sqrt{\frac{\hbar}{2m\omega}}, \qquad \Delta P_\alpha = \sqrt{\frac{m\hbar\omega}{2}}, \qquad \Delta X_\alpha \Delta P_\alpha = \frac{\hbar}{2}.}$$

(g) Derive the position-space form of the coherent state.

Calculation

One elegant way is to solve

$$(\hat{a} - \alpha)\phi_\alpha(x) = 0$$

in the position representation.

In x-space,

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\, x + \sqrt{\frac{\hbar}{2m\omega}}\frac{d}{dx}.$$

Therefore the equation becomes

$$\left(\sqrt{\frac{m\omega}{2\hbar}}\, x + \sqrt{\frac{\hbar}{2m\omega}}\frac{d}{dx} - \alpha\right)\phi_\alpha(x) = 0.$$

Multiply by $\sqrt{\frac{2m\omega}{\hbar}}$:

$$\left(\frac{m\omega}{\hbar}x + \frac{d}{dx} - \sqrt{\frac{2m\omega}{\hbar}}\alpha\right)\phi_\alpha(x) = 0.$$

Hence

$$\frac{d\phi_\alpha}{dx} = \left(\sqrt{\frac{2m\omega}{\hbar}}\alpha - \frac{m\omega}{\hbar}x\right)\phi_\alpha.$$

Integrating,

$$\phi_\alpha(x) = C\exp\left(\sqrt{\frac{2m\omega}{\hbar}}\alpha\, x - \frac{m\omega}{2\hbar}x^2\right).$$

Writing $\alpha = \Re(\alpha) + i\Im(\alpha)$ and completing the square gives the Gaussian wave packet form

$$\phi_\alpha(x) = e^{i\theta_\alpha}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\exp\left[-\frac{(x-\langle X\rangle_\alpha)^2}{4(\Delta X_\alpha)^2} + \frac{i}{\hbar}\langle P\rangle_\alpha x\right],$$

with

$$\langle X\rangle_\alpha = \sqrt{\frac{2\hbar}{m\omega}}\Re(\alpha), \qquad \langle P\rangle_\alpha = \sqrt{2m\hbar\omega}\Im(\alpha), \qquad \Delta X_\alpha = \sqrt{\frac{\hbar}{2m\omega}}.$$

Therefore

$$|\phi_\alpha(x)|^2 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\exp\left[-\frac{(x-\langle X\rangle_\alpha)^2}{2(\Delta X_\alpha)^2}\right].$$

This is a Gaussian wave packet centered at $\langle X\rangle_\alpha$ with width ΔX_α.

Answer

The coherent state in position representation is

$$\boxed{\phi_\alpha(x) = e^{i\theta_\alpha}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\exp\left[-\frac{(x-\langle X\rangle_\alpha)^2}{4(\Delta X_\alpha)^2} + \frac{i}{\hbar}\langle P\rangle_\alpha x\right]}$$

and its probability density is a Gaussian packet:

$$\boxed{|\phi_\alpha(x)|^2 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\exp\left[-\frac{(x-\langle X\rangle_\alpha)^2}{2(\Delta X_\alpha)^2}\right].}$$

Section 4: Bonus — time evolution of coherent states At time $t = 0$ the oscillator is prepared in the coherent state

$$\psi(x,0) = \phi_{\alpha_0}(x).$$

Study the time evolution under

$$\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right).$$

(a) Show that

$$\psi_t = e^{-i\hat{H}t/\hbar}\psi_0 = e^{-i\omega t/2}\phi_{\alpha_t}, \qquad \alpha_t = \alpha_0 e^{-i\omega t}.$$

Calculation

Expand the initial state in the energy eigenbasis:

$$\phi_{\alpha_0} = e^{-|\alpha_0|^2/2} \sum_{n=0}^{\infty} \frac{\alpha_0^n}{\sqrt{n!}} \varphi_n.$$

The time-evolution operator acts as

$$e^{-i\hat{H}t/\hbar} \varphi_n = e^{-i\omega(n+1/2)t} \varphi_n.$$

Therefore

$$\begin{aligned} \psi_t &= e^{-|\alpha_0|^2/2} \sum_{n=0}^{\infty} \frac{\alpha_0^n}{\sqrt{n!}} e^{-i\omega(n+1/2)t} \varphi_n \\ &= e^{-i\omega t/2} e^{-|\alpha_0|^2/2} \sum_{n=0}^{\infty} \frac{(\alpha_0 e^{-i\omega t})^n}{\sqrt{n!}} \varphi_n. \end{aligned}$$

Since

$$|\alpha_0 e^{-i\omega t}| = |\alpha_0|,$$

this is exactly

$$\psi_t = e^{-i\omega t/2} \phi_{\alpha_t}, \qquad \alpha_t = \alpha_0 e^{-i\omega t}.$$

Answer

$$\boxed{\psi_t = e^{-i\omega t/2} \phi_{\alpha_0 e^{-i\omega t}}.}$$

Thus a coherent state remains coherent under time evolution of the harmonic oscillator.

(b) Compute the expectation values of position, momentum, and energy, and compare them to classical mechanics.

Calculation

Since at time t the state is again a coherent state with parameter

$$\alpha_t = \alpha_0 e^{-i\omega t},$$

the formulas from the previous section apply directly:

$$\langle X \rangle_t = \sqrt{\frac{2\hbar}{m\omega}} \Re(\alpha_t), \qquad \langle P \rangle_t = \sqrt{2m\hbar\omega} \Im(\alpha_t).$$

Write

$$\alpha_0 = |\alpha_0| e^{i\varphi}.$$

Then

$$\alpha_t = |\alpha_0| e^{i(\varphi - \omega t)}.$$

Hence

$$\boxed{\langle X \rangle_t = \sqrt{\frac{2\hbar}{m\omega}}\, |\alpha_0| \cos(\omega t - \varphi)}$$

and

$$\boxed{\langle P \rangle_t = -\sqrt{2m\hbar\omega}\, |\alpha_0| \sin(\omega t - \varphi).}$$

These are exactly the classical solutions for a harmonic oscillator:

$$x_{\mathrm{cl}}(t) = A\cos(\omega t - \varphi), \qquad p_{\mathrm{cl}}(t) = m\dot{x}_{\mathrm{cl}}(t),$$

with amplitude

$$A = \sqrt{\frac{2\hbar}{m\omega}}\, |\alpha_0|.$$

The energy expectation value is

$$\langle H \rangle_t = \hbar\omega \left(|\alpha_t|^2 + \frac{1}{2} \right).$$

Since $|\alpha_t| = |\alpha_0|$, this is constant:

$$\boxed{\langle H \rangle_t = \hbar\omega \left(|\alpha_0|^2 + \frac{1}{2} \right).}$$

The classical harmonic oscillator energy is also conserved. Thus the expectation values of X and P follow the classical trajectory exactly, while the energy remains constant.

Answer

$$\boxed{\langle X \rangle_t = \sqrt{\frac{2\hbar}{m\omega}} \Re(\alpha_0 e^{-i\omega t}), \qquad \langle P \rangle_t = \sqrt{2m\hbar\omega} \Im(\alpha_0 e^{-i\omega t}),}$$

and

$$\boxed{\langle H \rangle_t = \hbar\omega \left(|\alpha_0|^2 + \frac{1}{2} \right).}$$

The position and momentum expectation values follow the classical harmonic motion exactly.

(c) Compute the uncertainties ΔX_t and ΔP_t.

Solution

Since ψ_t is again a coherent state for all t, the uncertainties remain those of any

coherent state:

$$\Delta X_t = \sqrt{\frac{\hbar}{2m\omega}}, \qquad \Delta P_t = \sqrt{\frac{m\hbar\omega}{2}}.$$

Therefore

$$\Delta X_t \, \Delta P_t = \frac{\hbar}{2}$$

at all times.

So the wave packet oscillates without changing shape: its center moves classically, while its width stays constant.

Answer

$$\boxed{\Delta X_t = \sqrt{\frac{\hbar}{2m\omega}}, \qquad \Delta P_t = \sqrt{\frac{m\hbar\omega}{2}}.}$$

The coherent state remains a minimum-uncertainty wave packet at all times.

Additional Notes / Side Calculations

Side Calculation

Useful identities for the harmonic oscillator:

$$\begin{aligned}
\hat{a}\varphi_n &= \sqrt{n}\,\varphi_{n-1}, \\
\hat{a}^\dagger\varphi_n &= \sqrt{n+1}\,\varphi_{n+1}, \\
\hat{N} &= \hat{a}^\dagger\hat{a}, \\
X &= \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}+\hat{a}^\dagger), \\
P &= -i\sqrt{\frac{m\hbar\omega}{2}}(\hat{a}-\hat{a}^\dagger).
\end{aligned}$$

Useful Gaussian integral:

$$\frac{1}{\pi}\int_{\mathbb{C}} d^2\alpha\, e^{-|\alpha|^2}\alpha^n(\alpha^*)^m = n!\,\delta_{nm}.$$

This is the key identity behind the completeness relation of coherent states.

Notes

Physical interpretation:

- The Bernoulli distribution models a binary random event; its fluctuations are maximal at $p = 1/2$.
- For the Poisson distribution, mean and variance coincide, which is a characteristic signature of counting statistics.
- The Gaussian has vanishing odd central moments because it is symmetric about its

mean.

- In a constant electric field, the harmonic oscillator potential is merely shifted in space; the shape and level spacing are unchanged.
- Coherent states are the most classical oscillator states: the packet center follows the classical orbit, while the quantum uncertainty stays minimal and constant.